

Sahil Yadav

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Experience

EPICS Initiative: Daan Bridge

VIT Bhopal University

Full Stack Developer & Software Engineering Lead

Jan 2025 – Present

- Spearheaded the backend development of a scalable service-oriented architecture for a 5-member cross-functional team using React.js, Node.js, and MySQL, streamlining decentralized donation workflows.
- Implemented a secure JWT-based authentication protocol and granular Role-Based Access Control (RBAC), securing **100%** of administrative gateways against unauthorized vector injections.
- Automated an integrated regression testing pipeline via GitHub Actions to validate donation and payment API routes, directly reducing manual deployment overhead by **45%**.

Technical Projects

VulnShield — DevSecOps Vulnerability Scanner | Python, Flask, PostgreSQL, Docker, Nmap, OWASP ZAP

- Architected a vulnerability assessment engine wrapping Nmap socket scripts and OWASP ZAP daemons inside a microservice-ready Flask REST API framework.
- Formulated a comprehensive test suite via PyTest to evaluate CVSS risk-scoring metrics, successfully achieving **81% code coverage** across 12+ critical endpoints.
- Orchestrated a containerized CI/CD pipeline using GitHub Actions, guaranteeing zero-downtime dependency linting, Docker layer building, and remote repository safety checks.

SecurePixel — Image Steganography Pipeline | Python, Flask, NumPy, Huffman Coding, LSB, Spread Spectrum

- Programmed an accelerated cryptographic pipeline merging Huffman trees with Spread Spectrum LSB algorithms to embed encrypted payloads without causing visible image degradation.
- Resolved critical character overflow failures by writing 22 structural regression unit tests in PyTest, successfully hitting **86% automated code coverage**.
- Constructed an automated coverage metrics reporter inside the continuous testing pipeline, validating end-to-end steganography operations across REST API boundaries.

Technical Skills

Languages: C++, Python, Java (JDK 17), JavaScript (ES6+), SQL.

Security & AppSec: OWASP Top 10, JWT Authentication, Vulnerability Assessment, Network Scanning, Risk Scoring.

Backend & Cloud: Node.js, Express.js, Flask, AWS (EC2, S3), RESTful APIs, Microservices.

Databases: MySQL, PostgreSQL (Index Optimization, Relational Architectures).

Tools & OS: Git, GitHub Actions, Nmap, OWASP ZAP, Docker, NumPy, Pandas, Linux.

Education

VIT Bhopal University

Bhopal, India

Bachelor of Technology in Computer Science Engineering (Cybersecurity) | **CGPA: 8.99/10.0** Sep 2023 – Jun 2027

Research Publications

Daan Bridge: An Integrated System for Donations, Volunteering, and Disaster Response — IEEE WorldSUAS 2026 (IEEE Xplore).

Collaborated within a 5-author research team to publish a scalable systems engineering paper; formulated layered service-oriented architectural layouts and automated task queues to prevent system downtime.

Certifications

AWS Certified Solutions Architect – Associate — Amazon Web Services.

IBM Cybersecurity Analyst Professional — IBM SkillsBuild.

Blockchain and Cryptocurrency (Elite Certification) — NPTEL / IIT Madras.

Achievements

Competitive Programming: Achieved a **1480** contest rating on LeetCode. Solved **257 DSA problems** exclusively in C++, demonstrating advanced algorithmic proficiency by clearing **174 Medium and Hard** challenges.

Algorithmic Consistency: Awarded the **100 Days Badge (2026)** on LeetCode. Built strong competency in complex optimization patterns including Dynamic Programming and Backtracking.